

PRAGYANSH CHATURVEDI

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OBJECTIVE

Software engineer with **2 years of experience** working with **linux and open source software** (debugging and maintaining Ubuntu for the Raspberry Pi). Interested in **operating systems, low-level programming and optimizations**, embedded computers and SBCs, networking and security

EDUCATION

Bachelor's of Technology (Computer Science), Indian Institute of Technology Roorkee July 2024

Cumulative Grade: 9.432/10

Top 10 in a class of 120 CS students

*Awarded the **Shri MA Rao Convocation award** for 2nd best all round performance in the class of 2024*

EXPERIENCE

Software Engineer

August 2024 - Current

Canonical

Remote, India

- Part of the **Ubuntu Foundations - Architectures** squad, where I work on **maintaining Ubuntu** (an OS that runs on **100,000,000+ devices**) and hardware enablement for Raspberry Pi devices. Involved in **debugging and updating 2000+ of packages in Ubuntu** and the linux kernel to move Ubuntu forward. [Link to my LP profile](#)
- **Leading the effort** on improving the desktop multimedia experience by **profiling and optimizing open source software** like ffmpeg, pulseaudio, libmutter, libsdl and the linux kernel for Ubuntu 26.04
- Worked on **failsafe boot mechanism** and **robot-automated hardware image testing** for the Raspberry Pi in Ubuntu 25.10 - resulting in the **reduction of 350 hours of dev effort yearly**
- Leveraged libcamera, rpCAM-apps and picamera to **enable the Raspberry Pi camera stack** on Ubuntu 25.04 and achieve parity with Raspbian - **benefitting 1000+ end users**
- Part of **Ubuntu Sponsors and Ubuntu MOTU**, one of 70 people (independent from Canonical) who **review and make decisions on all changes to Ubuntu**
- Earned a **fast track promotion** to Prof I level (equivalent to Google L4)

Software Engineering Intern

June 2024

TurboML

Remote, India

- TurboML is a real-time streaming ML platform.
- **Profiled and fixed performance bottlenecks** in the inference pipeline of TurboML's streaming models. Further improved performance by implementing the [LR algorithm for wait-free reads](#)
- **Created a custom RPC implementation in C++17** for the project to give a unified "actions/" endpoint to all models and actionable objects

Summer@EPFL Intern

July 2023 - November 2023

École Polytechnique Fédérale de Lausanne

Lausanne, Switzerland

- Worked in the Robust Scalable System Software Lab under Prof. Sanidhya Kashyap, mentored by Lyu Tao
- **Wrote a linux kernel fuzzer** to find security vulnerabilities in the eBPF verifier of the linux kernel in C and Go by **generating valid random eBPF programs using eBPF bytecode**
- [This](#) is a **list of bugs found by the fuzzer** and reported to the LKML

Quantitative Research Intern

May 2023 - July 2023

JP Morgan

Mumbai, India

- Part of QR Rates team, worked on the pricing of Interest Rate Swaps and other PRIIPS
- **Automated a rate prediction pipeline** from the derivative inputs to the final report being published to the stakeholders
- Contributed organised, quality and rigorously tested Python code to the codebase, which included a **Monte Carlo Simulation engine, a connector library to a distributed DB, a connector to a reactive circuit**, REST endpoints and load balancing, along with tests, all in a pythonic way

Contributor - ScummVM

Google Summer of Code

May 2022 - Sep 2022

Remote

- ScummVM is a cross-platform collection of engines for various retro games
- **Extended Director engine support to Macromedia Director 4.0** movies by implementing required systems and fixing bugs. This **enabled 100+ more games** on ScummVM — [Link to work](#)

PROJECTS

PythonBPF — [Link](#)

Sep 2025 - Present

- PythonBPF is a **compiler for writing eBPF kernelspace code in pure Python**
- A subset of Python is allowed (inspired from FastAPI and PyTorch's syntax) to write the eBPF code. Currently supports some basic map types and helper functions. Some BCC examples and their enhanced demos are available in the github repo
- This project has **190 stars** on GitHub and was presented at **LPC 2025** — [Link to talk](#)

GBEmu — SDS Labs — [Link](#)

Jan 2023 - Jun 2023

- GBEmu is a Nintendo GameBoy emulator written in C++ using SDL2 for graphics, input and audio
- **Created the CPU, MemoryMap, Interrupt Handler, Timers and the GPU** of the system. Is able to run **200+ GB games** (MBC0 and MBC1)

ErDOS — SDS Labs — [Link](#)

Dec 2022 - Mar 2023

- ErDOS is a **math problem solving website** (similar to [projecteuler](#) but for the IITR campus)
- **Led the auth migration to auth0** including DB and rate limiting. Stressed the website for a **peak load of 10000+ submissions per minute**. **Achieved this within a week**, just in time for hosting Endgame 9.0

Rootex — SDS Labs — [Link](#)

Oct 2021 - Dec 2022

- Rootex is an advanced C++ 3D game engine based on the Entity-Component-System architecture and uses a Lua scripting Engine for making games
- **Created and profiled the in-development game with Rootex, fixing bugs in the engine**, and optimizing it for the game

CONFERENCES

- **FOSDEM 2026** — Speak on the progress of **PythonBPF** in the eBPF devroom. [Link to proposal](#)
- **Linux Plumbers Conference 2025** — Presented **PythonBPF** at LPC 2025, a **peer-selected premier conference** for engineers and researchers working around the linux kernel. **One of 250** accepted speakers — [Link to talk](#)
- **Ubuntu Summits and Canonical Sprints** — Gave 5 minute lightning talks on **hardware debugging over SWD** and PythonBPF — [Link](#)

ACHIEVEMENTS

- **eBPF Summit 2025** — Earned a **special mention** at the eBPF Summit 2025 hackathon, where the judges acknowledged the potential of PythonBPF to **positively impact the eBPF ecosystem**
- **Segfault compiler hackathon 2025** — Runner-up at the hackathon held at Indian Institute of Science during IICT, where we presented an early prototype of PythonBPF. [Link](#)
- **CSAW ESC Finals 2022** — Part of Team Volga which ranked **2nd in CSAW ESC Finals based off ML Security** challenges organized by New York University
- **CSAW CTF Finals 2021** — Part of Team SDS Labs which ranked **2nd in India and 6th in the world** among UG teams which participated in the CSAW CTF Finals organized by New York University
- **IAO OCSC 2017** — **One of 25 students from all over India** who attended the Orientation cum Selection Camp for International Astronomy Olympiad 2017

SKILLS

Languages	C, C++, Python, Rust, Go, Bash, Perl
Software Packages	CMake, Meson, GDB, Valgrind, eBPF, perf, vtune, Docker, Git, Jenkins
Language Skills	English (SRW), Hindi (SRW), French (RW)